

Cloud Computing Lab 5 – Assignment 1

AWS RDS PostgreSQL Integration with Redmine on EC2

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Class	T.E. Computer Engineering – COMPS B	Subject	Cloud Computing
Academic Year	2026–2027	RDS	redmine-lab5-db

1. Objective

The objective was to deploy an Amazon RDS PostgreSQL database and integrate it with the Redmine application running on the Lab 4 Amazon EC2 instance. The work demonstrates managed database deployment, secure EC2-to-RDS connectivity, application operation, and CRUD operations through Redmine.

2. Architecture

Internet / Client Browser	Amazon EC2 – redmine-lab4 Nginx : 80 → Redmine / Puma : 127.0.0.1:3000 EC2 Security Group: TCP 80	Amazon RDS – PostgreSQL redmine-lab5-db Database: lab5db Port: 5432 SSL Private database access
HTTP : 80	EC2 → RDS PostgreSQL : 5432 (SSL)	RDS Security Group TCP 5432 from EC2 SG only

The client accesses Redmine through HTTP port 80. Nginx forwards requests to Redmine/Puma on the EC2 instance. Redmine connects privately to the RDS PostgreSQL endpoint on port 5432. The RDS security group restricts database ingress to the EC2 security group.

3. Database Deployment and Connectivity Evidence

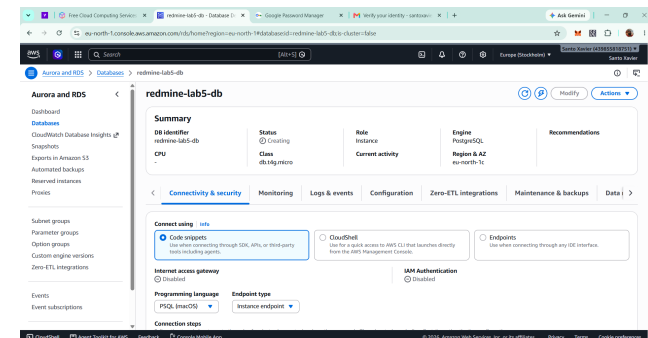


Figure 1 – Amazon RDS PostgreSQL instance

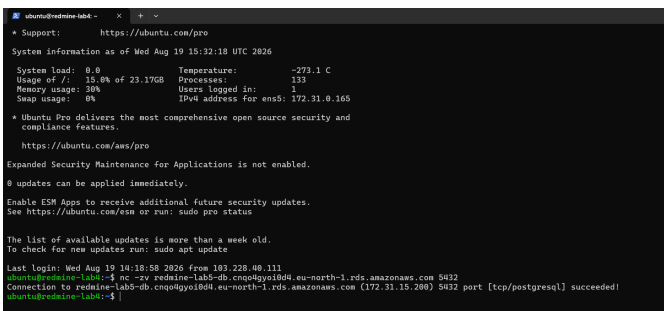


Figure 2 – EC2-to-RDS TCP 5432 connectivity test

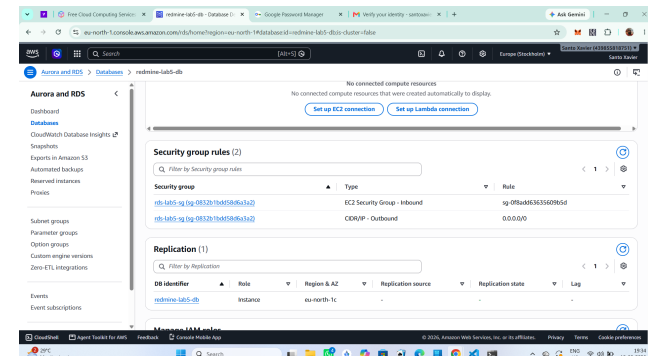


Figure 3 – RDS security-group configuration

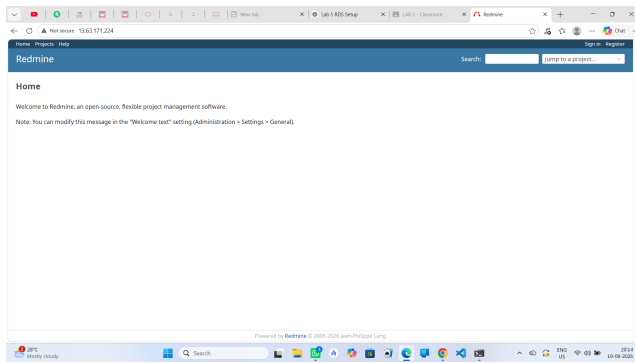


Figure 4 – Redmine application running on EC2

4. DB and Security Settings

• Engine: PostgreSQL • DB identifier: **redmine-lab5-db** • Database: **lab5db** • Port: **5432** • Region: Europe (Stockholm) • EC2: **redmine-lab4**.

Security was implemented using AWS Security Groups. PostgreSQL access is restricted to the EC2 security group rather than exposing port 5432 to the public internet. The database is used by the EC2-hosted Redmine application and is not intended for direct external access.

5. CRUD Operations Through the EC2-Hosted Redmine Application

CRUD functionality was demonstrated through the Redmine web application. The evidence below shows successful create, read, update, and delete actions in the application backed by the RDS database.

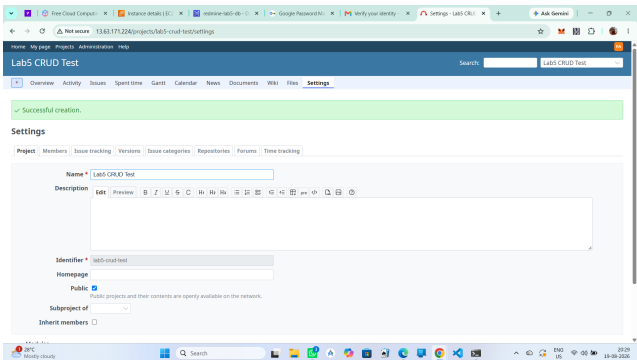


Figure 5 – CREATE: project/record creation completed successfully

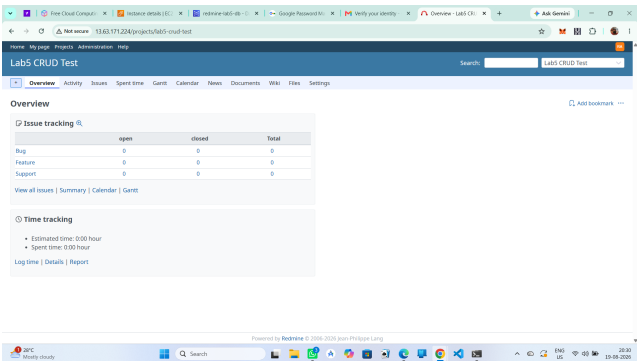


Figure 6 – READ: project/record visible in Redmine

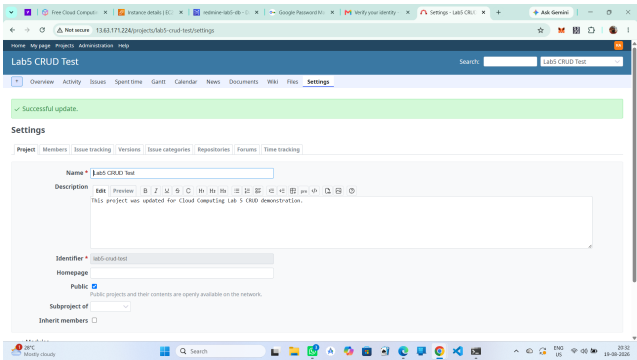


Figure 7 – UPDATE: project information updated successfully

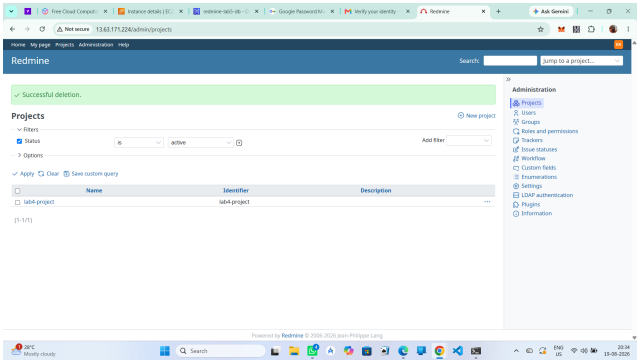


Figure 8 – DELETE: deletion completed successfully

6. Application and Repository Details

EC2 Application URL	http://13.63.171.224/
GitHub Repository	https://github.com/SAnto-spec/cloud-computing-lab5-rds-redmine-10699
Application	Redmine on Amazon EC2 using Nginx/Puma
Database	Amazon RDS for PostgreSQL

7. Conclusion

The Lab 5 assignment was successfully completed by deploying PostgreSQL on Amazon RDS and integrating it with the Redmine application hosted on the existing EC2 instance. EC2-to-RDS connectivity was verified on PostgreSQL port 5432, database access was

restricted using a security-group rule, and the application successfully performed Create, Read, Update, and Delete operations. This demonstrates a secure cloud architecture in which the application server and managed database are separated while communicating over the AWS private network.

Result: AWS RDS PostgreSQL integration with Redmine on EC2 and mandatory CRUD demonstration completed successfully.